responsible for the perception of light in photophysiological process (induction and inhibition of flowering) is called **Phytochrome**. Besides photoperiod certain plants require a low temperature in the earlier stages for flowering. Many biennial and perennial plants are induced to flower by low temperature (0°C to 5°C). This process is called **vernalization** and the reversal effect of vernalization is called **devernalization**. The condition of a seed when it fails to germinate even in suitable environmental condition is called **seed dormancy**. Thus, dormancy can be overcome by following methods such as scarification, impaction, stratification, alternating temperatures and light. Senescence refers to all collective, progressive and deteriorative processes which ultimately lead to complete loss of organization and function. Senescence is of four types and they are overall, top, deciduous and progressive. Senescence is controlled by plant's own genetic programme. Death of the plant or its parts consequent to senescence is called **Programmed Cell Death** (PCD). The final stage of senescence is abscission. Abscission is a physiological process of shedding of organs from the parent plant body. The study of functioning of plants under adverse environmental conditions is called **stress physiology**. The environmental stress may broadly be divided into biotic and abiotic stress. The adverse effect on plants caused by other living organisms such as viruses, bacteria, fungi, parasites, insects, weeds is competing plants are called biotic stress. Abiotic stress may occur due to an atmospheric condition or soil condition.

Evaluation

1. Select the wrong statement from the following:
   a. Formative phase of the cells retain the capability of cell division.
   b. In elongation phase development of central vacuole takes place.
   c. In maturation phase thickening and differentiation takes place.
   d. In maturation phase, the cells grow further.

2. If the diameter of the pulley is 12 inches, length of pointer is 10 inches and distance travelled by pointer is 5 inches. Calculate the actual growth in length of plant.
   a. 3 inches b. 6 inches
   c. 12 inches d. 30 inches
3. _____ is the powerful growth inhibitor
   a. Ethanol b. Cytokinins
   c. ABA d. Auxin
4. Select the correctly matched one

A) Human urine	i) Auxin –B
B) Corn gram oil	ii) GA_3
C) Fungus	iii) Abscisic acid II
D) Herring fish sperm	iv) Kinitin
E) Unripe maize grains	v) Auxin A
F) Young cotton bolls	vi) Zeatin

   a) A-iii, B-iv, C-v, D-vi, E-i, F-ii,
   b) A-v, B-i, C-ii, D-iv, E-vi, F-iii,
   c) A-iii, B-v, C-vi, D-i, E-ii, F-iv,
   d) A-ii, B-iii, C-v, D-vi, E-iv, F-i
5. Seed dormancy allows the plants to
   a. overcome unfavourable climatic conditions
   b. develop healthy seeds
   c. reduce viability
   d. prevent deterioration of seeds
6. Which one of the following method are used to break the seed dormancy?
   a) Scarification b) Impaction
   c) Stratification d) All the above.
7. Write the physiological effects of Cytokinins.
8. Describe the mechanism of photoperiodic induction of flowering.
9. Give a brief account on Programmed Cell Death (PCD)
10. What are the physiological effects of plants facing drought condition?
11. Explain the mechanism of biotic stress.

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